Yanshuai Qin

Germany | +4915258525488 | qinyshuai@gmail.com | LinkedIn | GitHub | Website

Profile

Machine Learning Researcher with a PhD in Mathematics from UC Berkeley and postdoctoral research experience. Built end-to-end LLM and quantitative ML pipelines spanning model training, fine-tuning, distributed GPU training, post-training, quantization, evaluation, optimization, and deployment. Strong Python, machine learning, mathematical, and research background.

Technical Skills

- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, LLM pretraining, supervised fine-tuning (SFT), RLHF, quantization, model evaluation, deployment
- **Programming:** Python, C++, Git/GitHub, Linux, Bash
- **Data & Scientific Computing:** NumPy, pandas, SciPy, Jupyter, XGBoost, A/B testing
- **Mathematics:** Probability, statistics, numerical analysis, optimization
- **Languages:** Chinese (native), English (fluent)

Machine Learning Projects

Nanochat End-to-End LLM Pipeline — GitHub

- Built and validated an end-to-end LLM pipeline on Kaggle using the nanochat codebase, covering tokenizer training, pretraining, supervised fine-tuning, distillation, DPO/PPO/RL post-training, quantization, evaluation, and deployment.
- Adapted distributed training to run on 2× T4 GPUs by debugging distributed execution, stabilizing data loading in notebook environments, and producing working checkpoints and evaluation outputs at each stage.

Cost-Aware Limit Order Book Forecasting — GitHub

- Developed an end-to-end pipeline for short-horizon limit-order-book forecasting and benchmarked XGBoost, CNN-LSTM, and Transformer models using leakage-safe walk-forward validation.
- Built a transaction-cost-aware backtesting framework incorporating spread, fees, slippage, calibration, and latency.

Experience

Postdoctoral Researcher Nov 2023 - Dec 2025

University of Regensburg, Germany — Supervisor: Moritz Kerz

- Jointly with a mathematician from IMJ-PRG, solved a difficult problem posed by Skorobogatov and Zarhin a decade earlier, using Python and SageMath for computational exploration and verification; invited to speak at EPFL and IRMA, and selected to present the work at the 2025 SRI in Algebraic Geometry, a leading once-a-decade conference.
- Collaborated with an IBM cryptography scientist and an expert in p-adic methods to generalize the Artin-Grothendieck theorem (1968) to arbitrary dimension, building on a collaborator's breakthrough on the 30-year-old Berthelot conjecture; used Python and deep learning to study arithmetic statistics of abelian varieties. The work was presented by the collaborator at an IHES conference.

Visiting Scholar Oct 2023 - Nov 2023

Beijing International Center for Mathematical Research, Beijing

Graduate Student Instructor and Researcher Aug 2018 - Sep 2023

University of California, Berkeley

- Taught numerical analysis/optimization, probability, statistics, and differential geometry; developed Python-based coding assignments and numerical labs for students, including implementations of core numerical algorithms.
- Conducted research in arithmetic geometry using Python and deep learning for symbolic and numerical computation, resulting in 5 publications and multiple invited talks at international conferences and institutes, including PRIMA Congress 2022.

Education

PhD, Mathematics — University of California, Berkeley (Advisor: Xinyi Yuan) 2018 - 2023

MSc, Mathematics — Nanjing University 2016 - 2018

BSc, Mathematics — Nanjing University 2011 - 2015

Selected Publications & Preprints

- **On p-torsions of geometric Brauer groups** (with Zhenghui Li). *Journal für die reine und angewandte Mathematik (Crelles Journal)* (2026). DOI; arXiv:2406.19518.
- **On the Brauer groups of fibrations**. *Mathematische Zeitschrift* 307, Article 18 (2024). DOI.
- **F-isocrystals of Higher Direct Images of p-divisible Groups** (with Zhenghui Li). Submitted. arXiv:2506.11736.
- **On the Brauer groups of fibrations II**. Submitted. arXiv:2103.06910.
- **On geometric Brauer groups and Tate-Shafarevich groups**. arXiv:2012.01681.
- **Comparison of different Tate conjectures** (with Veronika Ertl and Timo Keller). Submitted. arXiv:2012.01337.

Selected Invited Talks and Conference Presentations

- Summer Research Institute in Algebraic Geometry, Fort Collins, CO — contributed talk, Jul 28, 2025.
- Bernoulli Center, EPFL — Workshop on Arithmetic Geometry of K3 Surfaces, May 8, 2025.
- IRMA, University of Strasbourg — Arithmetic and Algebraic Geometry Seminar, Dec 19, 2024.
- PRIMA Congress 2022 — Special Session in Arithmetic Geometry, Dec 9, 2022.
- Berkeley-Tokyo Workshop on Number Theory and Arithmetic Geometry, Mar 11, 2022.

Awards

- Lehmer Fellowship in Number Theory (Spring 2022); UC Berkeley Summer Grants (2020-2023).
- UC Berkeley Conference Travel Grant.